

Solar Panels

Category 1

Since I have a digital meter and a home battery, I am interested in knowing how many watt-hours my solar panels produce during certain periods of the day, week, ... My app currently only shows that figure per hour, but I also want to know (as an example) how many watt-hours I received during every 4-hour period: between 0h and 4h, between 1h and 5h, and so on. I want to add that piece of software to my app, but in such a way that the user can configure things like the length of the period and the time span over which the data should be processed. That would also make it useful for similar applications, such as water or gas consumption, how much time I waste on social media, ...

Task

The general idea of this assignment is that you are given a sequence of numbers, for example the yield of my solar panels over a number of consecutive hours during a given period. Then, for each hour, you calculate the sum of those values over the given period and output all those sums. The length of the period can also vary and is included in the input.

Input

The first line gives the number of test cases. For each test case, three lines follow:

- The first line gives the length $L > 0$ of the period over which the sums must be calculated.
- The second line contains an integer $D > 0$, indicating how many hours/days of data will follow.
- The third line contains D positive integers representing the measurement results of those D hours/days.

Example Input

```
3
4
9
1 2 3 4 5 6 7 8 9
3
9
9 8 7 6 5 4 3 2 1
1
5
1 2 3 4 5
```

Output

For each test case, print one line: the line begins with the case number (the first test case gets number 1). Then, for each of the $(D - L + 1)$ periods, output the sum of the measured values in that period. Numbers on the same line must be separated by a single space.

Example Output

1	10	14	18	22	26	30	
2	24	21	18	15	12	9	6
3	1	2	3	4	5		